User Guide for ACTS

ACTS is a test generation tool for constructing *t*-way combinatorial test sets. Currently, it supports t-way test set generation with t ranging from 1 to 6. Combinatorial testing has been shown very effective in detecting faults that are caused by unexpected interactions between different contributing factors. The tool provides both command line and GUI interfaces.

This document is organized as follows. Section 1 provides an overview of the core features of ACTS. Section 2 provides information about the command line interface. Section 3 provides information about the GUI interface.

1 Core Features

1.1 T-Way Test Set Generation

This is the core feature of ACTS. A system is specified by a set of parameters and their values. A test set is a *t*-way test set if it satisfies the following property: Given *any t* parameters (out of all the parameters) of a system, every combination of values of these *t* parameters is covered in at least one test in the test set.

Currently, ACTS supports *t*-way test set generation for $1 \leq t \leq 6$. Empirical studies show that t being up to 6 is sufficient for most practical applications. A special form of 1-way testing, called base-choice testing, is implemented in ACTS. Base-choice testing requires that every parameter value be covered at least once and in a test in which all the other values are base choices. Each parameter has one or more values designated as base choices. Informally, base choices are “more important” values, e.g., default values, or values that are used most often in operation.

Several test generation algorithms are implemented in ACTS. These algorithms include IPOG, IPOG-D, IPOG-F, IPOG-F2. In general, IPOG, IPOG-F, and IPOG-F2 work best for systems of moderate size (less than 20 parameters and 10 values per parameter on average), while IPOG-D is preferred for larger systems.

ACTS supports two test generation modes, namely, *scratch* and *extend*. The former allows a test set to be built from scratch, whereas the latter allows a test set to be built by extending an existing test set. In the *extend* mode, an existing test set can be a test set that is generated by ACTS, but is incomplete because of some newly added parameters and values, or because of a test set that is supplied by the user and imported into ACTS. Extending an existing test set can save earlier effort that has already been spent in the testing process.

1.2 Mixed Strength

This feature allows different parameter groups to be created and covered with different strengths. For example, consider a system consisting of 10 parameters, P1, P2, …, and P10. The first relation can be created that consists of all the parameters with strength 1 or 2. Then, additional relations can be created if some parameters are believed to have a higher degree of interaction, based on the user's domain knowledge. For instance, a relation could be created that consists of P2, P4, P5, P7 with strength 4 if the four parameters are closely related to each other, and their 4-way interactions could trigger certain software faults. ACTS allows arbitrary parameter relations to be created, where different relations may overlap or subsume each other. In the latter case, relations that are subsumed by other relations will be ignored by the test generation engine. Mixed strength is only supported in IPOG and IPOG-F algorithm.

1.3 Constraint Support

Some combinations are not valid from the domain semantics, and must be excluded from the resulting test set. For example, when we want to make sure a web application can run in different Internet browsers and on different Operating Systems, the configuration of IE on Mac OS is not a valid combination. A test that contains an invalid combination will be rejected by the system (if adequate input validation is performed) or may cause the system to fail. In either case, the test will not be executed properly, which may compromise test coverage, if some (valid) combinations are only covered by this test.

ACTS allows the user to specify constraints that combinations must satisfy to be valid. The specified constraints will be taken into account during test generation so that the resulting test set will cover, and only cover, combinations that satisfy these constraints. Currently, constraint support is only available for the IPOG and IPOG-F algorithm. Constraint support for other algorithms will be added in a future release.

1.4 Coverage Verification

This feature is used to verify whether a test set satisfies t-way coverage, i.e. whether it covers all the t-way combinations. A test set to be verified can be a test set generated by ACTS or a test set supplied by the user (and then imported into ACTS).

2. Command Line Interface

There is a jar file for both command line mode and GUI mode. In this section, we assume that the jar file is named **acts.jar**. (The actual jar files names in the release package may be different.) The command line mode can be executed using the

following command[1]:

java <options> -jar acts.jar <input_filename> [output_filename]

Alternatively, the command line mode can also be executed by using classpath:

java <options> -cp acts.jar edu.uta.cse.fireeye.console.ActsConsoleManager <input_filename> [output_filename]

The various options are:

-Dalgo=ipog|ipog_d|ipof|ipof2|basechoice|null

- ipog - use algorithm IPO (default)
- ipog_d - use algorithm IPO + Binary Construction (for large systems)
- ipof - use the IPOF method
- ipof2 - use the IPOF2 method
- basechoice - use Base Choice method
- null - use to check coverage only (no test generation)

-Ddoi=<int>

- specify the degree of interactions to be covered. Use -1 for mixed strength.

-Doutput=numeric|nist|csv|excel

- numeric - output test set in numeric format
- nist - output test set in NIST format (default)
- csv - output test set in CSV format
- excel - output test set in EXCEL format

-Dmode=scratch|extend

- scratch - generate tests from scratch (default)
- extend - extend from an existing test set

-Dchandler=no|solver|forbiddentuples

- no - ignore all constraints
- solver - handle constraints using CSP solver
- forbiddentuples - handle constraints using minimum forbidden tuples (default)

-Dcheck=on|off

- on - verify coverage after test generation
- off - do not verify coverage (default)

[1] The previous version command has been deprecated, which is changed as follow: java <options> -jar acts~~_cmd~~.jar ~~cmd~~ <input_filename> [output_filename], since acts_cmd.jar and acts_gui.jar have been replaced by a single acts.jar file.

-Dprogress=on|off

- on - display progress information (default)
- off - do not display progress information

-Ddebug=on|off

- on - display debug info
- off - do not display debug info (default)

-Drandstar=on|off

- on - randomize don't care values (default)
- off - do not randomize don't care values

-Dcombine=<all>

- all - every possible combination of parameters

The above usage information can be displayed using the following command:

java -jar acts.jar -h

In the command line, <input_file> contains the configuration information of the system to be tested. The format of a configuration file is illustrated using the following example:

```
[System]
-- specify system name
Name: TCAS

[Parameter]
-- general syntax is parameter_name (type) : value1, value2, ...
Cur_Vertical_Sep (int) : 299, 300, 601
High_Confidence (boolean) : TRUE, FALSE
Two_of_Three_Reports_Valid (boolean) : TRUE, FALSE
Own_Tracked_Alt (int) : 1, 2
Other_Tracked_Alt (int) : 1, 2
Own_Tracked_Alt_Rate (int) : 600, 601
Alt_Layer_Value (int) : 0, 1, 2, 3
Up_Separation (int) : 0, 399, 400, 499, 500, 639, 640, 739, 740, 840
Down_Separation (int) : 0, 399, 400, 499, 500, 639, 640, 739, 740, 840
Other_RAC (enum) : NO_INTENT, DO_NOT_CLIMB, DO_NOT_DESCEND
Other_Capability (enum) : TCAS_TA, OTHER
Climb_Inhibit (boolean) : TRUE, FALSE
```

```
[Relation]
-- this section is optional
-- general format Rx : (p1, p2, ..., pk, Strength)
R1 : (Cur_Vertical_Sep, Up_Separation, Down_Separation, 3)

[Constraint]
-- this section is also optional
Cur_Vertical_Sep != 299 => Other_Capability != "OTHER"
Climb_Inhibit = true => Up_Separation > 399

[Test Set]
-- set existing test set for extend mode. this is also optional
-- * represents don't-care value
Cur_Vertical_Sep, Other_Tracked_Alt, Alt_Layer_Value, Climb_Inhibit
299,1,0,false
```

Currently, three parameter types are supported: enum, bool (boolean), and int (integer). Note that, bool/enum types are **case insensitive** in parameter and constraint to avoid duplicated values; int type supports **negative values** in parameter and constraint now. Lines beginning with -- represents comments that exist only to improve the readability of the configuration file.

The default heap size for the Java Virtual Machine may not be adequate for large configurations. The user is recommended to change the default heap size, if necessary, using the following command:

java -Xms <initial heap size> -Xmx <max heap size> <options> -jar acts.jar <input_file> [output_file]

3 GUI Interface

There are two ways to launch the GUI front end. One way is to double-click the jar file, which is executable. The other way is to execute the jar file on the command prompt as follows:

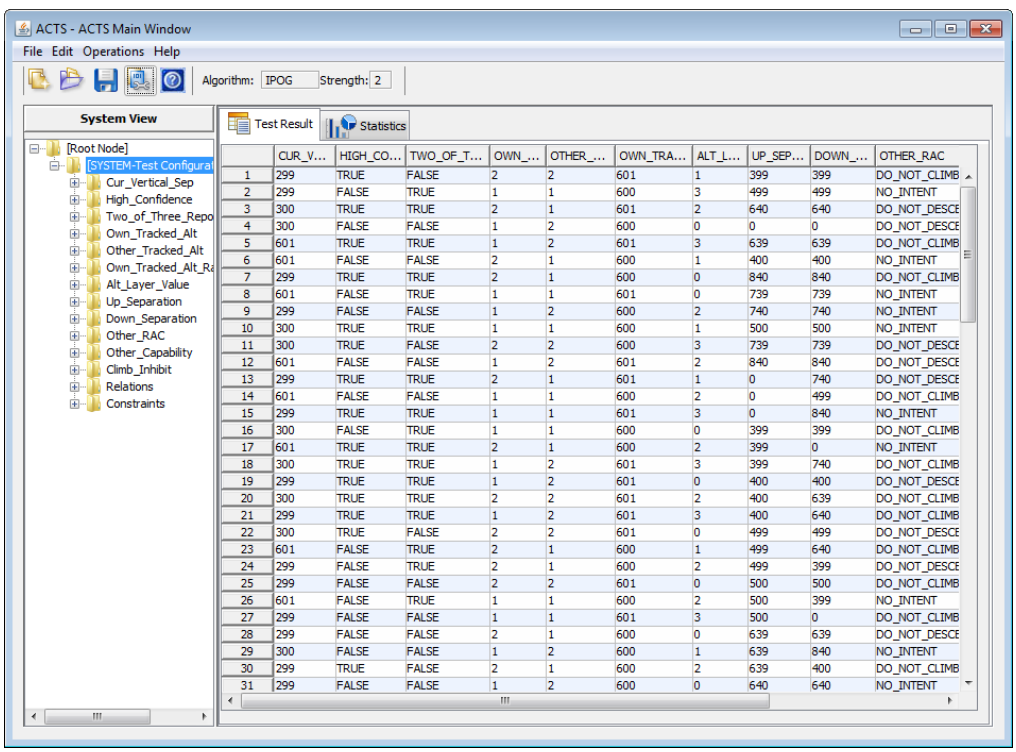
java -jar acts.jar

The following command can be used to change the default heap size for java virtual machine, if necessary:

java -Xms <initial heap size> -Xmx <max heap size> <options> -jar acts.jar

Figs. 1 and 2 show the general layout of the ACTS GUI. The System View component is a tree structure that shows the configurations of the systems that are currently open in the GUI. In the tree structure, each system is shown as a three-level hierarchy. That is, each system (top level) consists of a set of parameters (second level), each of which has a set of values (leaf level). If a system has relations and constraints, they will be shown in the same level as the parameters.

Right to the System View is a tabbed pane consisting of two tabs, namely, Test Result, which is shown in Fig. 1, and Statistics, which is shown in Fig. 2. The Test Result shows a test set of the currently selected system, where each row represents a test, and each column represents a parameter. Output parameters are also displayed as columns. The Statistics tab displays some statistical information about the test set. In particular, it includes a graph that plots the growth rate of the test coverage with respect to the tests in the test set displayed in the Test Result tab. Drawing the graph may involve expensive computations, and thus the graph is shown only on demand, i.e. when the Graph button is clicked.



	CUR_V...	HIGH_CO...	TWO_OF_T...	OWN_...	OTHER_...	OWN_TRA...	ALT_L...	UP_SEP...	DOWN_...	OTHER_RAC
1	299	TRUE	FALSE	2	2	601	1	399	399	DO_NOT_CLIMB
2	299	FALSE	TRUE	1	1	600	3	499	499	NO_INTENT
3	300	TRUE	TRUE	2	1	601	2	640	640	DO_NOT_DESCE
4	300	FALSE	FALSE	1	2	600	0	0	0	DO_NOT_DESCE
5	601	TRUE	TRUE	1	2	601	3	639	639	DO_NOT_CLIMB
6	601	FALSE	FALSE	2	1	600	1	400	400	NO_INTENT
7	299	TRUE	TRUE	2	1	600	0	840	840	DO_NOT_CLIMB
8	601	FALSE	TRUE	1	1	601	0	739	739	NO_INTENT
9	299	FALSE	FALSE	1	2	600	2	740	740	NO_INTENT
10	300	TRUE	TRUE	1	1	600	1	500	500	NO_INTENT
11	300	TRUE	FALSE	2	2	600	3	739	739	DO_NOT_DESCE
12	601	FALSE	FALSE	1	2	601	2	840	840	DO_NOT_DESCE
13	299	TRUE	TRUE	2	1	601	1	0	740	DO_NOT_DESCE
14	601	FALSE	FALSE	1	1	600	2	0	499	DO_NOT_CLIMB
15	299	TRUE	TRUE	1	1	601	3	0	840	NO_INTENT
16	300	FALSE	TRUE	1	1	600	0	399	399	DO_NOT_CLIMB
17	601	TRUE	TRUE	2	1	600	2	399	0	NO_INTENT
18	300	TRUE	TRUE	1	2	601	3	399	740	DO_NOT_CLIMB
19	299	TRUE	TRUE	1	2	601	0	400	400	DO_NOT_DESCE
20	300	TRUE	TRUE	2	2	601	2	400	639	DO_NOT_CLIMB
21	299	TRUE	TRUE	1	2	601	3	400	640	DO_NOT_CLIMB
22	300	TRUE	FALSE	2	2	601	0	499	499	DO_NOT_DESCE
23	601	FALSE	TRUE	2	1	600	1	499	640	DO_NOT_CLIMB
24	299	FALSE	TRUE	2	1	600	2	499	399	DO_NOT_DESCE
25	299	FALSE	FALSE	2	2	601	0	500	500	DO_NOT_CLIMB
26	601	FALSE	TRUE	1	1	600	2	500	399	NO_INTENT
27	299	FALSE	FALSE	1	1	601	3	500	0	DO_NOT_CLIMB
28	299	FALSE	FALSE	2	1	600	0	639	639	DO_NOT_DESCE
29	300	FALSE	FALSE	1	2	600	1	639	840	NO_INTENT
30	299	TRUE	FALSE	2	1	600	2	639	400	DO_NOT_CLIMB
31	299	FALSE	FALSE	1	2	600	0	640	640	NO_INTENT

Figure 1. The Main Window - Test Result Tab

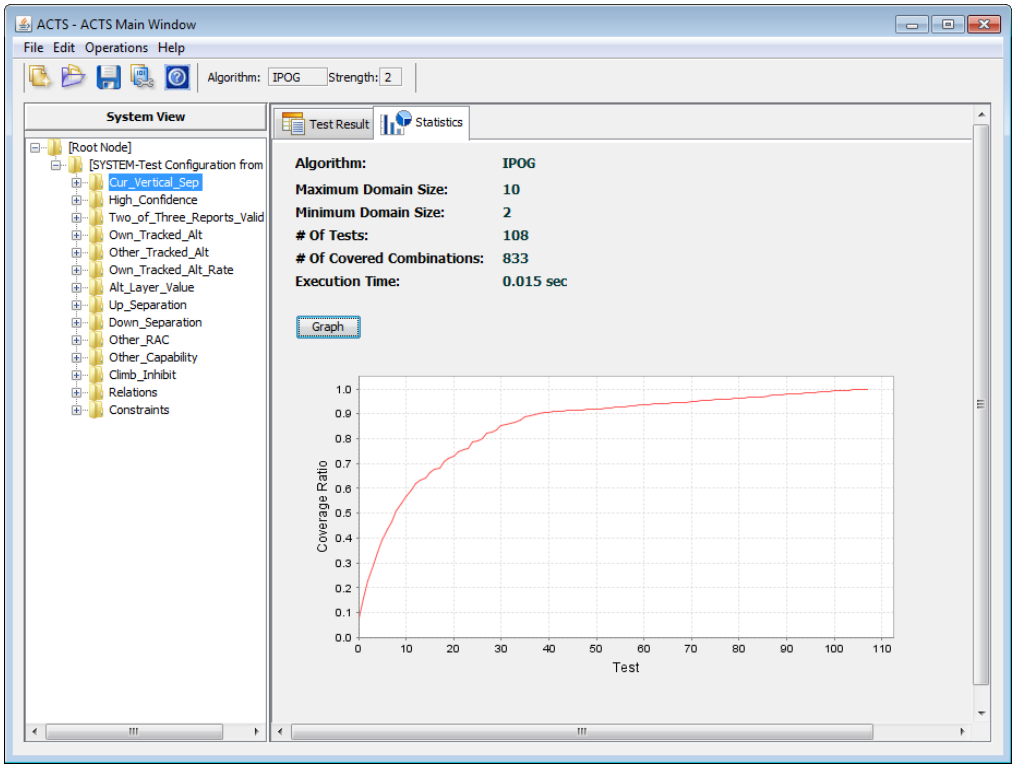


Figure 2. The Main Window - Statistics Tab

3.1 Create New System

To create a new system, select menu *System -> New*, or the first icon in the toolbar, to open the *New System* window. The *New System* window contains a tabbed pane of three tabs, namely, *Parameters*, *Constraints* and *Relations*. The three tabs are shown in Figs. 3, 4, and 5, respectively.

The *Parameters* tab (Fig. 3) allows the user to specify the parameters, as well as the values of those parameters, in the new system. Currently, four parameter types are supported, *Boolean*, *Enum*, *Number*, and *Range*. *Range* is a convenience feature that allows multiple, consecutive integers to be input quickly. Note that parameter names cannot contain spaces. (The characters that can be contained in a parameter name are the same as those in a variable name in Java programs.)

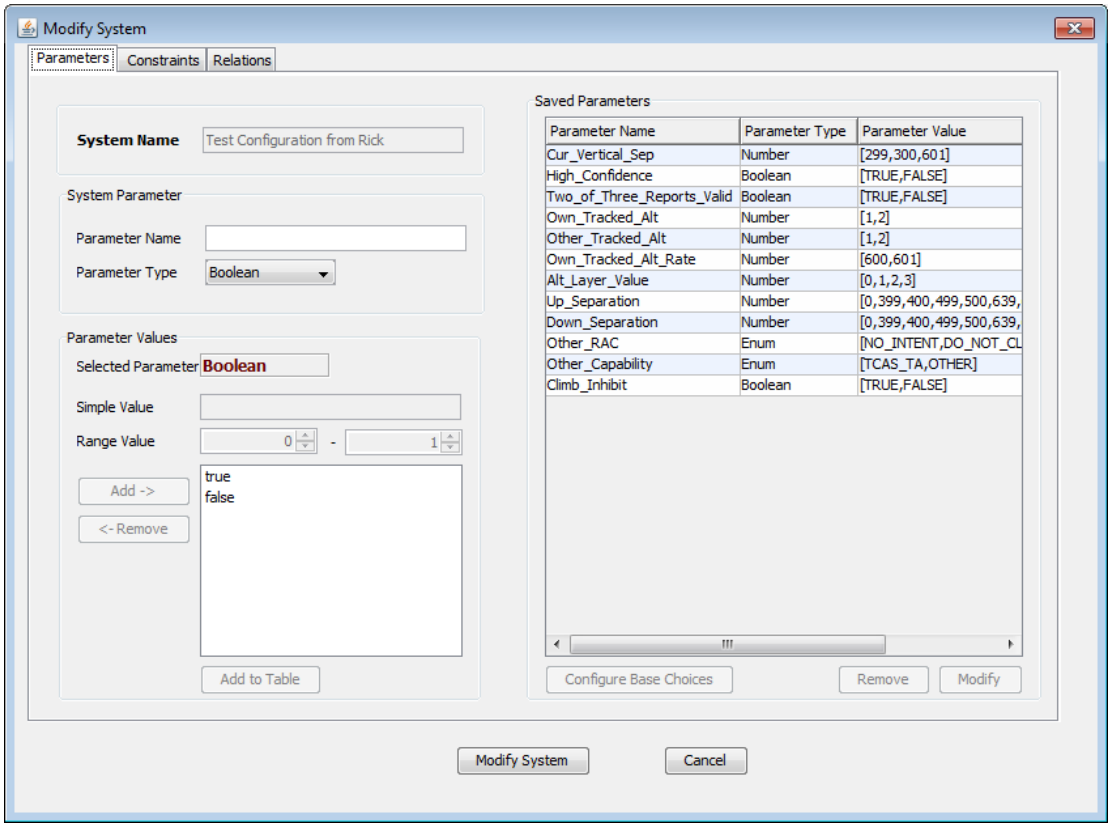


Figure 3. New System Window - Parameters

The *Constraints* tab (Fig. 4) allows the user to specify constraints so that invalid combinations can be excluded from the resulting test set. Generally speaking, a constraint is specified using a restricted form of first-order logical formulas. In the following, we give a formal syntax of the expressions that can be used to specify a constraint:

- <Constraint> ::= <Simple_Constraint> | <Constraint> <Boolean_Op> <Constraint>
- <Simple_Constraint> ::= <Term> <Relational_Op> <Term>
- <Term> ::= <Parameter> | <Parameter> <Arithmetic_Op> <Parameter> | <Parameter> <Arithmetic_Op> <Value>
- <Boolean_Op> ::= **&&, ||, =>**
- <Relational_Op> ::= **>, <, =, !=, >=, <=**
- <Arithmetic_Op> ::= **+, -, *, /, %**

There are three types of operators: (1) *Boolean* operators (*Boolean_Op*), including &&, ||, =>; (2) *Relational* operators (*Relational_Op*), including =, !=, >, <, >=, <=; and (3) *Arithmetic* operators (*Arithmetic_OP),* including +, -, *, /, %. Note that

arithmetic operators can appear in a term expression (<*Term*>) only if the parameters involved in the term expression are of type Number or Range. Also, four of the relational operators, namely, >, <, >=, <=, can appear in a simple constraint expression (*Simple_Constraint*) only if both of the terms involved in the simple constraint are evaluated to a parameter value of type Number or Range.

The following are examples of various constraints that can be specified:

Constraint 1: (OS = “Windows”) => (Browser = “IE” || Browser = “FireFox” || Browser = “Netscape”), where OS and Browser are two parameters of type Enum. This constraint specifies that if OS is Windows, then Browser has to be IE, FireFox, or Netscape.

Constraint 2: (P1 > 100) || (P2 > 100), where P1 and P2 are two parameters of type Number or Range. This constraint specifies that P1 or P2 must be greater than 100.

Constraint 3: (P1 > P2) => (P3 > P4), where P1, P2, P3, and P4 are parameters of type Number or Range. This constraint specifies that if P1 is greater than P2, then P3 must be greater than P4.

Constraint 4: (P1 = true || P2 >= 100) => (P3 = “ABC”), where P1 is a Boolean parameter, P2 is a parameter of type Number or Range, and P3 is of type Enum. This constraint specifies that if P1 is true or P2 is no less than 100, then P3 must be “ABC”.

A constraint can be directly typed in the Constraint Editor. The user is provided with the system configuration and the operators that can be used. The left hand side of the Constraint window displays the system configuration in a table format and the operators on the top of the system configuration table. Note that parameter values that are strings must be quoted in double quotes; otherwise, they will be considered as parameter names. The right bottom of this frame shows all the added constraints. By clicking the button “Load From File”, one may load constraints from a plain txt file, in which each line contains a constraint expression. For example, a constraint file (in TXT format) contains two lines:

```
Cur_Vertical_Sep != 299 => Other_Capability != "OTHER"
Climb_Inhibit = true => Up_Separation > 399
```

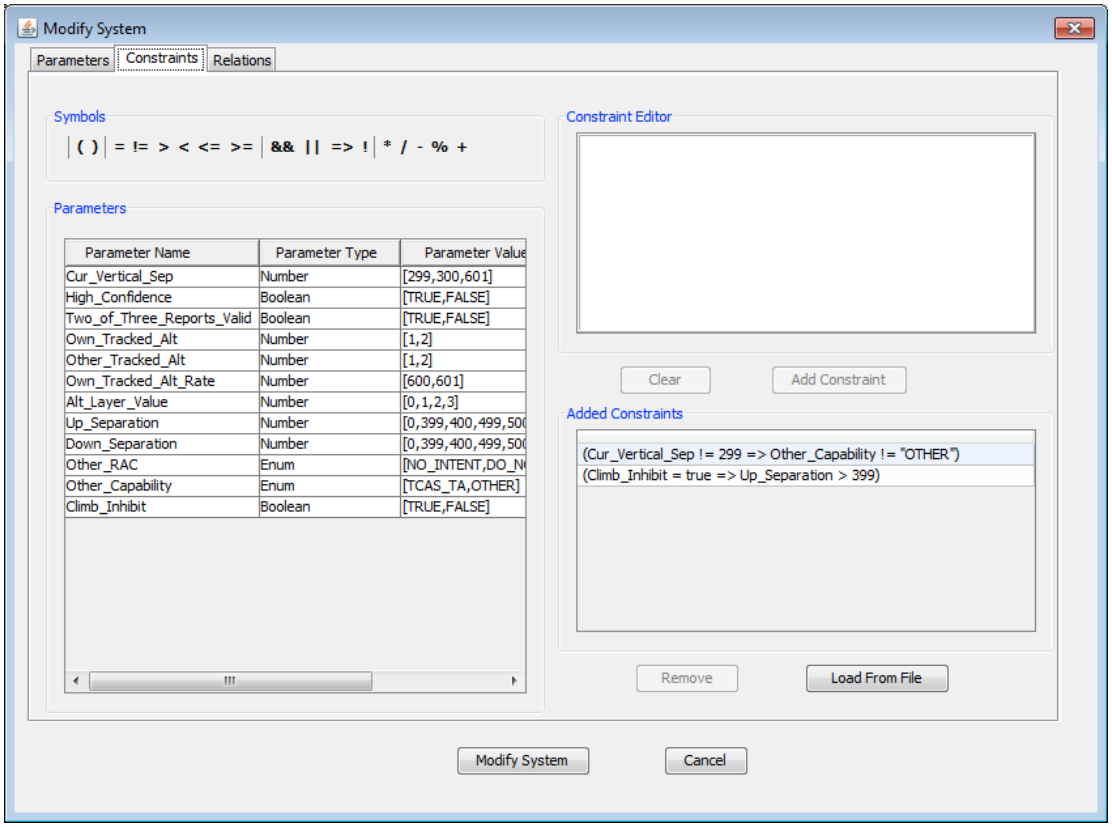


Figure 4. New System Window - Constraints

An existing constraint can be removed by selecting the constraint in the Added Constraint table and then clicking on the Remove button. Currently, ACTS does not allow an existing constraint to be directly edited. In order to edit an existing constraint, the user needs to remove the constraint first and then add the desired constraint as a new constraint.

The *Relations* tab (Fig.5) allows user to edit customized relations over different parameter groups. To generate using customized relations, the test strength (see Section 3.2) must be set to “Mixed Strengths”.

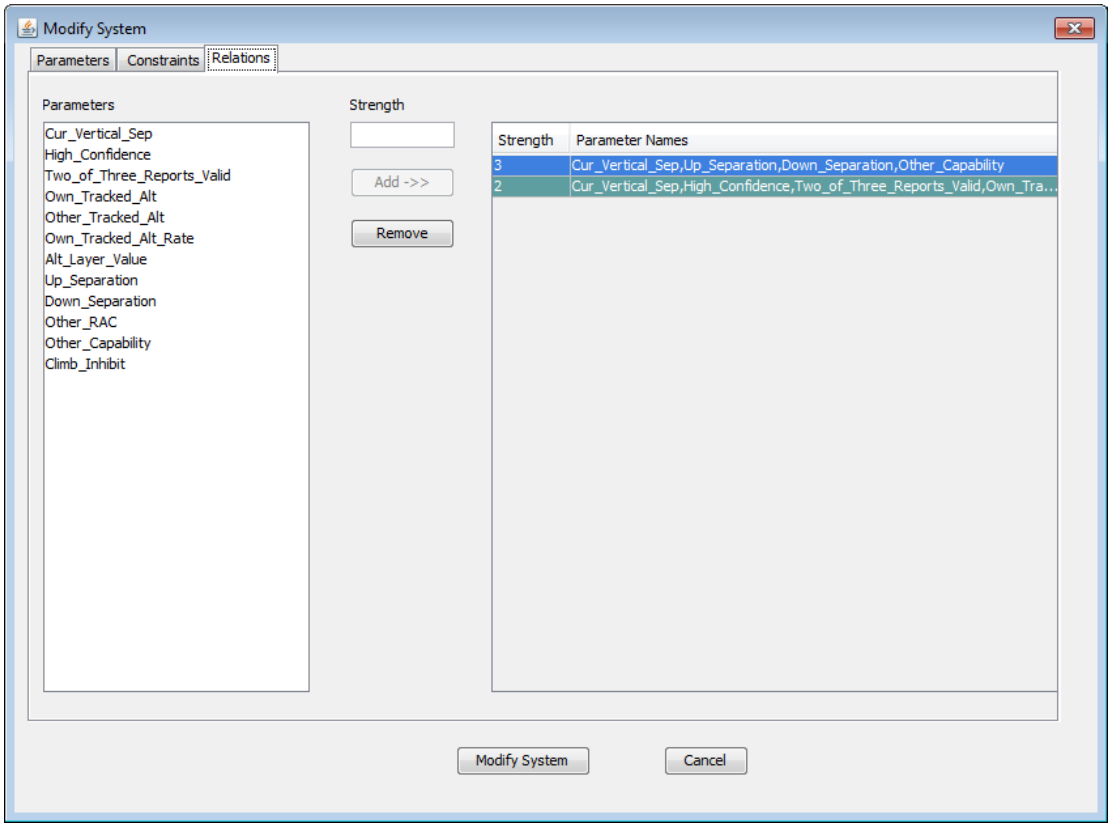


Figure 5. New System Window - Relation

3.2 Build Test Set

To build a test set for a system that is currently open, select the system in the System View, and then select menu *Operations -> Build*. The latter selection brings up the *Options* window, as shown in Fig. 6, which allows the following options to be specified for the build operation. Note that some options may not be supported by some algorithms. The IPOG algorithm is highly recommended.

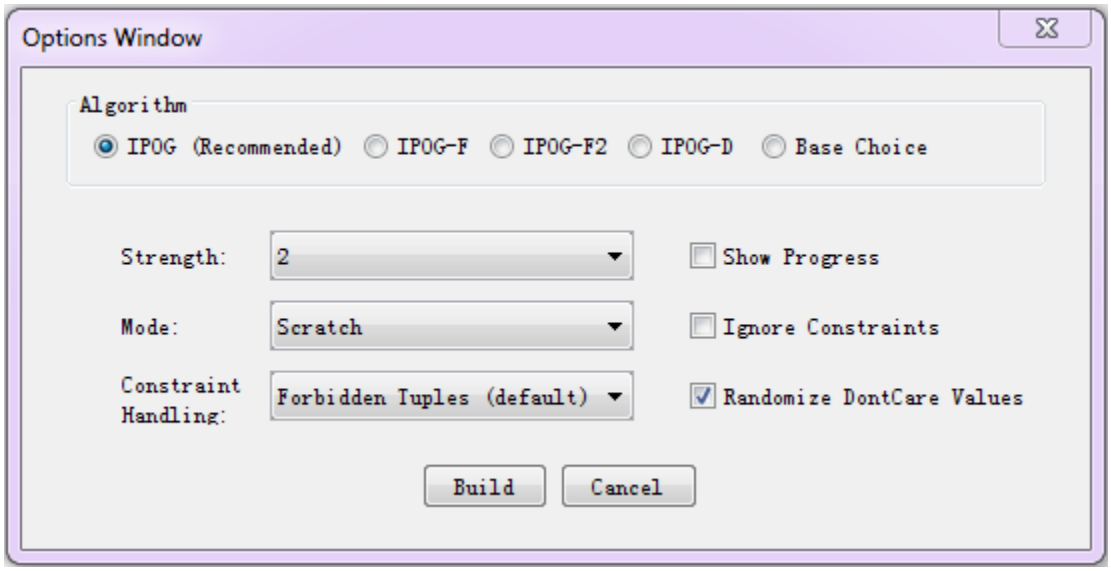


Figure 6. Build Options Window

- *Algorithm*: This option decides which algorithm to be used for test generation. By default, the IPOG algorithm is selected. As mentioned in Section 1.1, IPOG, IPOG-F, IPOG-F2 and work best for systems of moderate size, while IPOG-D is preferred for larger systems. The IPOG algorithm is recommended to use.
- *Randomize Don't Care Values*: If this option is checked, then all the *don't care* values (*) in the resulting test set will be replaced with a random value. The coverage will not be affected if we replace a *don't care* value with a real value.
- *Ignore Constraints*: If this option is checked, all constraints will be ignored. For some generation algorithms which don't support constraints, this option will be automatically checked.
- *Strength*: This option specifies the strength of the test set. Currently, ACTS supports a strength value ranging from 1 to 6. If the strength is set to a number between 1 and 6, only the specified strength will be used for test generation. If the strength is set to "Mixed Strengths", the customized relations in the system (Section 3.1, *Relations* tab) will be used. For Base Choice algorithm the strength will be set to 1.
- *Mode*: This option can be *Scratch* or *Extend*. The former specifies that a test set should be built from scratch; the latter specifies that a test set should be built by extending an existing test set (shown in the Test Result tab). Recall that the current test set in the system may not be complete as the system configuration may have changed after the last build or the test set may be imported from outside.
- *Constraint Handling*: This option can be *Forbidden Tuples* or *CSP Solver*.

The former specifies that the validity check is handled using minimum forbidden tuples, which are generated from constraints. Now, this is a completed feature and is set to be default option for user. The latter specifies that the validity check is handled by a CSP solver. The former works faster when given a system under test, which has a large number of parameters with many constraints, and strength $t \geq 2$. The latter works faster when given a system that has a small number of parameters with a few constraints and strength $t \leq 2$. Both options will not affect the resulting test set.

- *Progress*: If this option is turned on, progress information will be displayed in the console. Note that in order to obtain the console, the GUI must be started from a command prompt instead of by double-clicking the executable jar file.

After the build operation is completed, the resulting test set will be displayed in the *Test Result* tab of the *Main* window.

3.3 Modify System

To modify an existing system, select the system in the tree view, and then select menu *Edit -> Modify*. The *Modify System* window is the same as the *New System* window except that the name of the system cannot be changed. A parameter cannot be removed if it is involved in a relation or constraint. In this case, please remove the involved relation or constraint first.

A parameter can be added in the same way as during the New System operation. A parameter can be removed by selecting the parameter in the Saved Parameters table on the right hand side, and then clicking on the Remove button under the table.

The “Configure Base Choices” button is used to set base choices for each parameter. In the Configure Base Choices Window (Fig. 7), if the checkbox next to a parameter value is checked, this value is designated as a base choice value. A parameter may have multiple base choice values, meaning that multiple checkboxes can be checked. Base choice values are only used by the base-choice testing algorithm. The base choice values will be highlighted after they are added to the system. As mentioned earlier, base-choice testing is a special form of 1-way testing.

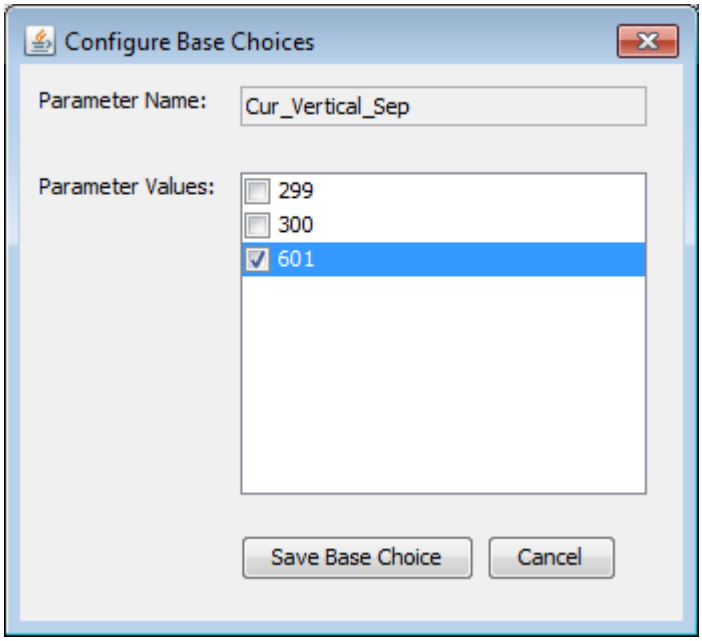


Figure 7. Configure Base Choices Window

The values of a parameter can be modified by selecting the parameter on the Saved Parameters table on the right hand side, and by clicking on the Modify button. An enum type parameter value cannot be removed if it is involved in a constraint. In this case, please remove the involved constraint first.

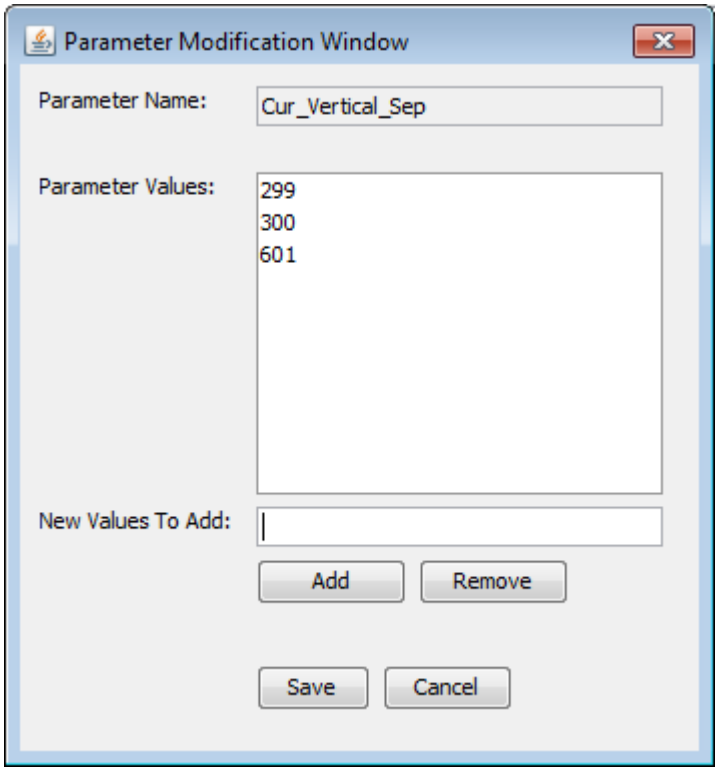


Figure 8. Parameter modification window.

3.4 Save/Save As/Open System

To save an existing system, select the system in the tree view, and then select menu *System -> Save* or *Save As*. When a newly created system is saved for the first time, or when *Save As* is select, a standard file dialog will be brought up, where the user can specify the name of the file to be saved. The system will display a confirmation window if the file to be saved already exists.

Note that now we support user to both open a system from a TXT file, and save the content in TXT format.

3.5 Import/Export Test Set

To import a test set of a system, the user must first create the system, in terms of adding its parameters and values into ACTS, as described in Section 3.1. Then, select menu *Operations -> Import*, and select the format of the file containing the test set. Currently, two file formats are supported: CSV-R, which stands for Comma Separated Values with Row headers, and CSV-RC, which stands for Comma Separated Values with Row and Column headers. (CSV-RC is mainly used to facilitate integration with Excel.) The following are two example files, one for each format:

CSV-R format:

```
P1,P2,P3,P4,P5
0,2,2,3,6
3,2,4,2,2
2,1,2,1,3
3,2,5,0,5
```

CSV-RC format:

```
,P1,P2,P3,P4,P5
Test1,0,2,2,3,6
Test2,3,2,4,2,2
Test3,2,1,2,1,3
Test4,3,2,5,0,5
```

The parameter values in each row must be separated by “,”. There can be arbitrary space between two values. After the file format is selected, a standard file selection window appears through which the user can browse through the system and select the file containing the test set to be imported.

To export a test set that exists in the GUI, first select the corresponding system so that the test set is displayed in the *Test Result* tab of the *Main* window, and then select

Operations -> Export. Currently, three formats are supported, namely, NIST Format, Excel Format and CSV Format. A snippet of an exported test set in the NIST format is shown below:

```
# ACTS Test Suite Generation: Tue May 19 06:45:46 CDT 2015
#  "(don't care)" represents don't care value
# Degree of interaction coverage: 2
# Number of parameters: 4
# Maximum number of values per parameter: 2
# Number of configurations: 6
------------Test Cases--------------

Configuration #1:

1 = P1=true
2 = P2=true
3 = P3=true
4 = P4=true

-------------------------------------

Configuration #2:

...
```

A snippet of an exported test set in the CSV Format is shown below:

```
# ACTS Test Suite Generation: Tue May 19 17:15:35 CDT 2015
#  "*" represents don't care value
# Degree of interaction coverage: 2
# Number of parameters: 4
# Maximum number of values per parameter: 2
# Number of configurations: 6
P1,P2,P3,P4
true,true,true,true
true,false,false,false
...
```

A snippet of an exported test set in the Excel Format is shown below:

```
# ACTS Test Suite Generation: Tue May 19 06:58:57 CDT 2015
#  "*" represents don't care value
# Degree of interaction coverage: 2
# Number of parameters: 4
# Maximum number of values per parameter: 2
# Number of configurations: 6
Test Case No.  P1      P2      P3      P4
1        true    true    true    true
2        true    false   false   false
...
```

3.6 Verify T-Way Coverage

This operation is typically used to verify the coverage of a test set that is imported from outside of ACTS. It will first find all possible combinations that need to cover, and then find how many combinations are actually covered by current test set. To verify *t*-way coverage of a test set, the user can select menu *Operations => Verify*, choose *With Constraints* or *Ignore Constraints*, and select a desired strength. Note that the number of possible combinations can be affected by two options. If choosing the "*Ignore Constraints*" menu, all constraints will not be considered. If the test strength is set to "*Mixed*", the customized relations (could be multiple relations) will be used.